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Project Title: Formula 1 Podium Finishers Predictor Using Race Result Data

Description:

Formula One (F1), which recently took place in Miami, is an annual event that uses a points system. I plan to build a model that predicts whether a Formula One driver will finish on the podium (1st, 2nd, or 3rd) in a given race.

Formula One results are affected by a variety of factors, including season, track, team, and starting grid position.

Data Collection:

I will use a Kaggle dataset containing historical results from **2019-2024, with** 128 unique races, 36 drivers, 22 teams, and 34 tracks.

Preprocessing will include handling missing values and encoding categorical columns (such as Driver, Team, and Track)

Planned Model:

I plan to start with a KNN classifier; however, I may compare results with Logistic Regression or a Decision Tree to assess performance metrics.

I will use an 80/20 train/test split, on results up to 2023, and use the 2024 results for external validation.

If the 2026 results become available, I can use these as additional external validation.